

ALL ABOUT CODING ROUND / SKILL TEST ROUND

✓ PHASE 1: Understand the Structure of a Coding Test

Most companies (especially product-based like Google, Amazon, Atlassian, etc., and startups) follow this pattern:

🧠 Typical Coding Test Format:

1. **Platform:** Hackerrank, Codility, HackerEarth, Mettl, CodeSignal, etc.
2. **Time:** 60–120 minutes
3. **Questions:**
 - a. 2 to 4 coding problems (DSA-based)
 - b. 0–1 SQL query or DBMS logic question
 - c. 0–2 MCQs on core CS subjects or CS theory
4. **Difficulty:** Mix of easy, medium, and hard
5. **Language:** You choose the language (Python/C++/Java)

💻 PHASE 2: What to Prepare (DSA + CS Essentials)

📅 DSA Topics (Ranked by Priority):

Priority	Topic	Subtopics to Master	Example Problems
★★★★	Arrays & Strings	Prefix sum, sliding window, 2 pointers	Kadane's Algo, Longest Substring
★★★★	Hashing & Maps	Frequency maps, sets	Two Sum, Group Anagrams
★★★★	Recursion & Backtracking	Subsets, permutations, N-Queens	Sudoku Solver, Combination Sum
★★★★	Linked List	Reversal, Cycle Detection, Merge Sort	Detect Loop, Reverse K nodes
★★★★	Stacks & Queues	Infix/Postfix, Monotonic Stack	Next Greater Element, Stock Span
★★★★	Trees & Binary Trees	DFS, BFS, LCA, Diameter, BST	Bottom View, Inorder Iterative

★ ★	Graphs	BFS, DFS, Union Find, Dijkstra, TopoSort	Number of Islands, Shortest Path
★ ★	Dynamic Programming	0/1 Knapsack, LCS, LIS, Matrix DP	House Robber, Edit Distance
★	Tries & Heaps	Trie insert/search, Min/Max heaps	Word Dictionary, Kth Largest Element

DSA INTERVIEW QNS

https://drive.google.com/file/d/1R7C6m_UWkp4VxgdO958XdQLBHCrJEvnj/view

MORE ABOUT DSA CHECK THIS OUT

<https://drive.google.com/file/d/1RNJxuXqg0KjEeCJnAmcJVk5rs4GKbYvJ/view?usp=drivesdk>

[krishnadev30/LeetCode-Questions-CompanyWise: Contains Company Wise Questions sorted based on Frequency and all time](#)

Core CS Subjects:

These are asked as either MCQs or short coding questions.

To prepare CORE CS subjects : check this out, it has all the things you need

https://drive.google.com/file/d/1OsloU1D_zkplEiAv-9-30MxldVsLg2Cd/view?usp=drive_link

Subject	Topics to Prepare	Must-Know Concepts
DBMS	Normalization, SQL Joins, Transactions	ACID, Primary/Foreign Key
OOPs	Inheritance, Polymorphism, Encapsulation	Real-life examples
OS	Process, Threads, Scheduling	FCFS, Round Robin, Mutex
CN	OSI vs TCP/IP, Protocols, IP, DNS, HTTP	Layers, Ports, IP classes

PHASE 3: Practice Plan (Smart & Structured)

Daily Coding Practice Routine:

- 1 easy + 1 medium + 1 hard (if time allows)
- Pick problems from: CHOOSE ONE AND FOLLOW IT TO THE END, MAIN THING IS DOING NOT FOLLOWING ALL ---YOU HAVE TO DO AT THE END OF DAY
 - [LeetCode DSA Sheet](#)
 - [Striver's DSA Sheet](#)
 - [Love Babbar 450 Sheet](#)

Mock Test Platforms:

Practice real timed tests at:

- [InterviewBit](#)
- [Codeforces](#)
- [Leetcode Weekly/Contest](#)
- [GFG Practice](#)

PHASE 4: Test Strategy (How to Attempt)

Time Management:

- Easy: 10–15 min
- Medium: 20–30 min
- Hard: Do last, if time allows

Smart Solving Tips:

- **Read all questions first**, start with what you're most confident in
- If stuck for 10 min, move to next
- Write **brute force** first, then optimize
- Always **use edge test cases** before submitting
- Comment your logic briefly if allowed

BONUS: Tools & Cheat Sheets

Tools:

- Whiteboard for dry runs
- VSCode for offline practice
- Online compilers: [replit.com](#), [ideone.com](#)

 Cheat Sheets:

- [DSA Templates \(by cp-algorithms\)](#)
- [Python DSA Cheatsheet](#)
- [Leetcode Pattern Sheet](#)

 Final Round Tip:

80% of candidates **fail because of lack of practice under pressure**. Not because they don't know the logic. Simulate 1-hour tests **3x a week** on Codeforces/LeetCode

NOTE: This document is a reflection of my personal journey, experiences from colleagues, friends, and many online resources like Google, YouTube, Reddit, and others, formatted with the help of AI to make it user-friendly and easy to read. I hope these insights help you on your coding journey!

ALL THE BEST